Midterm Exam CS 401/MCS 401 Computer Algorithms I Monday 13 July 2026

Open book exam. Submit your answers to gradescope.
In your explanations and justifications, write complete and grammatically correct sentences.
Please do not ask questions during the exam.

1. Given $(x_0, f_0), (x_1, f_1), \ldots, (x_n, f_n)$, $x_i \neq x_j$ for $i \neq j$, let $p(x) = \displaystyle\sum_{i=0}^{n} f_i \prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - x_i}{x_j - x_i}$.

   Express the operational cost to evaluate p at some x as a function f of n.
   Justify your formula for f explaining how you derived $f(n)$. /20

2. A directed graph $G = (V, E)$ of 6 vertices $V = \{0, 1, 2, 3, 4, 5\}$ has edges $E = \{(0,3), (5,0), (4,5), (5,1), (5,3), (0,1), (4,0), (3,1), (2,1), (2,4), (2,3), (2,5)\}$.

   (a) Apply the topological sorting algorithm to G. Show all steps.

   (b) What is a worst case input for the topological sort algorithm? Justify. /20

3. Consider the sequence $S = [a, b, c, d, b, b, e, d, c, c]$ for processing using a cache of size 2.

   (a) Show all steps in the farthest-in-the-future algorithm on S.

   (b) Use this example to illustrate that a nonreduced schedule does not lead to a number of cache misses fewer than the farthest-in-the-future eviction schedule. /15

4. Let $G = (V, E)$ be a weighted graph with $V = \{0, 1, 2, 3, 4, 5\}$ and edges $E = \{((0,2), 9), ((0,1), 9), ((0,4), 9), ((1,4), 6), ((2,5), 8), ((1,2), 7), ((2,3), 8), ((3,5), 2), ((4,5), 3), ((0,5), 9)\}$, where edge (i, j) with weight w is represented by $((i, j), w)$.

   (a) Construct a minimum spanning tree with Kruskal's algorithm.
   Show all steps in the evolution of the union-find data structure.

   (b) Explain how definition of the merge operation is important for the $O(m \log(m))$ cost for a graph with m edges. /15

5. Consider $p = x^3 - 9x^2 + 5x + 4$ and $q = 2x^3 + 4x^2 - 8x + 7$.

   (a) Use p and q to explain how to evaluate the product pq via divide and conquer at the roots of unity.

   (b) Justify the derivation of the recurrence relation for the cost.

6. Illustrate the application of divide and conquer to compute x^{256} for some number x.
   Assuming p is a power of 2, how many multiplications are needed to compute x^p?
   Justify your answer, referring to the example for $p = 256$. /15